

Now start your computer as usual, by pressing the power button on the front panel. When the PC runs, there will be 5V DC at the USB interface socket. As a result, relay RL2 energises via diode D6. The contacts of relay RL2 close switch S1 permanently, and LED2 glows continuously.

Release 'start' switch S1. Now your

PC manager is ready to use.

When you switch off your PC, relay RL2 de-energises. As a result, electric power from the switchboard (to which all peripherals are connected) is cut off. Switch S2 works here as an emergency bypass switch.

Assemble the circuit on a general-purpose PCB and enclose in a suitable

cabinet. Connect SOC1, SOC2 and USB socket along with switches S1 and S2 and LEDs (LED1 and LED2) on the front panel of the cabinet. Refer Fig. 2 for connections.

EFY note. Take care during fabrication and testing, as the circuit is at mains potential and may give you lethal shock. ●

HYDRO ALARM

■ PRADEEP G.

Here is a simple, low-cost circuit that sounds an alarm to alert you that your water tank is full. The circuit is built around transistors and introduces a squarewave signal into water via probes. The probes last longer as there is no electrolysis and no corrosion of the probes.

The entire circuit consists of three main sections: astable multivibrator, amplifier and switching.

The astable multivibrator outputs audio frequency in the form of square-

wave signal at the collector of transistor T2, which is fed to one of the probes fitted at the top of the water tank. The other probe is connected to the amplifier T3 via capacitor C5.

When the water tank is full, the audio frequency generated by the astable multivibrator spreads in water through the probes fitted at the top of the tank. The audio frequency available in water is received by capacitor C5, amplified by transistor T3 and fed to the switching section via capacitor C6. In the switching section, the amplified signal is fed to transistors T4 and

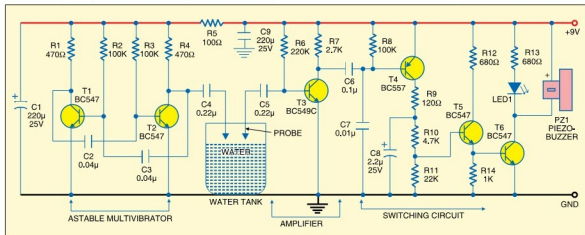
T5. The high output of the switching section drives transistor T6 to produce the sound from the piezobuzzer and light up LED1.

When the water tank is empty, neither the buzzer sounds, nor LED1 glows.

In brief, when the probes are shorted via water or any conductive liquid, the buzzer sounds.

Assemble the circuit on a general-purpose PCB and enclose in a suitable

cabinet. Use a conductive thick wire for probes and fix these probe wires at the top of the water tank such that when the water tank is full, both the probes touch the water level. ●



SMART BATTERY CHARGER

■ T.A. BABU

The life expectancy of a battery dramatically reduces when it is charged above the recom-

mended battery voltage. So the battery must be disconnected before over-charging happens. This disconnect function requires a battery-voltage sensing circuit.

Here is a simple charger circuit built around shunt regulator TL431

